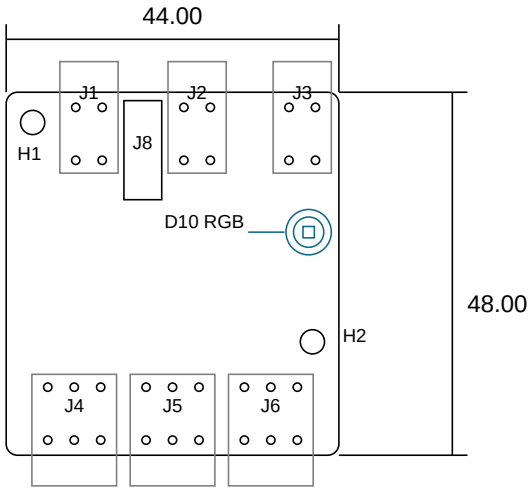
SC12 - nominal mechanical drawing

Rev H / component-side view / 1:1 at 100% print / dimensions in mm



PCB: 44 x 48 x 1.6 mm; corner radius 1.5 mm. Four layers, 1 oz copper.

Mounts: H1 (3.5, 4), H2 (40.5, 33); diameter 3.2 NPTH.

Keep hardware contact diameter <=6.4 mm; use insulating supports as needed.

Conservative terminal envelope: 44 x 56.1 mm, before wires/button clearance.

Terminal height: 8.5 above mounting face; pin tips 3.5 below that face.

On a 1.6 mm PCB, pin tips project nominally 1.9 mm beyond the bottom.

79 underside SMT parts: keep enclosure clear of both board faces.

D10: center (40, 18.5), height 0.65; 6 mm circular area reserved.

Printed duct starting size: outside diameter <=6; inside diameter 4.

Duct lower face: 0.85-1.15 above PCB; no contact with LED or screw boss.

J8 female socket: 8.5 above PCB; tails about 1.6 below bottom; allow cable access.

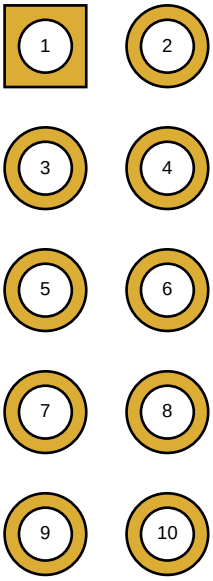
Housing geometry is nominal CAD. Physical and optical fit untested.

Origin: top-left PCB corner. X right; Y down in this drawing.

Service header and voltage jumpers

Component-side view / enlarged, not 1:1 / no stencil paste on these contacts

J8: recessed 2x5 / 2.54 mm socket; custom mapping



1 NRST
2 UART RX
3 UART TX
4 SWCLK
5 SWDIO
6 +3V3 VTref sense only
7 GND
8 GND
9 BOOT0
10 GND

Power through J1; probe power disconnected. Left column: odd pins; right column: even pins.

JP101 POWER

2 mm cap: 1-2 = 5V; 2-3 = 12V; no cap = off

Input mode: select NPN / PNP / push-pull in software

JP201 POWER

2 mm cap: 1-2 = 5V; 2-3 = 12V; no cap = off

Input mode: select NPN / PNP / push-pull in software

JP301 POWER

2 mm cap: 1-2 = 5V; 2-3 = 12V; no cap = off

Input mode: select NPN / PNP / push-pull in software

ONE CAP PER POWER HEADER. Joining all three pins shorts the rails.

JP2: bridge for 120 ohm CAN end termination; open at intermediate nodes.

User fits J8 and power headers/caps. Input bias is electronic. Change power caps with power off.